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Technical Paper

Convolutional neural network prediction of the particle size distribution of soil from close-range images

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Abstract

Accurate soil particle size distributions are essential for various geotechnical applications. In this study, we propose a convolutional neural network approach for predicting the particle size distribution using soil image analysis. Our model is trained on a diverse dataset of soil samples ranging from clayey silt to gravel. We employed transfer learning by using MobileNet pre-trained on ImageNet and adding additional layers to fine-tune the model for our specific task. The soil images were captured under standardised lab conditions using a dark chamber with constant lighting to ensure consistency. We implemented the model in Python and explored various neural network architectures, image resolutions and data augmentation techniques to optimise performance. The model predicts the particle size distribution through two parameters derived from the Weibull distribution. Our approach offers instantaneous predictions and demonstrates robustness across a wide range of soil types. We outperform previous studies by incorporating geotechnical classification and predicting the entire particle size distribution curve. Additionally, we applied explainable artificial intelligence techniques to enhance the transparency and interpretability of the model's predictions. Our findings highlight the effectiveness of the model and provide valuable insights into the relationship between soil image features and particle size characteristics.
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1. Introduction

In geotechnical engineering practice, soil classification is primarily based on the particle size distribution (PSD) curve for coarse soils (ISO 14688–1, 2017; ISO 17892-4, 2016). For fine soils, classification is often determined through assessments of plasticity (ISO 14688-2, 2017; ISO 17892-12, 2018). The PSD of mixed soils is typically determined in the laboratory using a combination of sieving for coarse-grained soils and sedimentation for fine-grained soils. This traditional method, while effective, is labor-intensive and time-consuming, which can be a significant limitation in practical applications (Alirezazadeh et al., 2021). The PSD is a crucial distribution in geotechnical engineering as it influences several soil properties such as residual water content (Esmaeelnejad et al., 2016), hydraulic conductivity in both saturated and unsaturated conditions (Hwang and Hong, 2006) and soil groutability (Vipulanandan and Ozgurel, 2009). Despite its importance, the conventional methods for determining PSD remain cumbersome, often requiring manual processing and prolonged analysis (Alirezazadeh et al., 2021).

In recent years, alternative methods have been developed to overcome these limitations. Techniques based on gamma-ray attenuation Oliveira et al. (1998), reflectance of visible and infrared radiation (Zuo et al., 2000) and image analysis have been explored as more efficient means of estimating PSD. However, these methods have their own limitations, such as issues with accuracy, applicability to a broad range of soil types, or high equipment costs. Among image-based methods, traditional image processing techniques have been used to classify soil textures.

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For example, Barman and Choudhury (2020) achieved an average accuracy of 91.37% in classifying soil images into 12 texture classes using hue, saturation and value histograms, colour moments, colour auto-correlograms, Gabor wavelets and discrete wavelet transforms. Similarly, Breul and Gourves (2006) used endoscopy images and spectral analysis to study the third-order moment's influence on PSD and Chung et al. (2010) developed a soil texture classification system using surface soil images, achieving 48% accuracy. While these methods demonstrated some success, they often struggled with limitations such as lower accuracy, inability to predict the full PSD curve or challenges in handling a diverse range of soil types. Recent advancements have seen the application of convolutional neural networks (CNNs), which are a class of artificial neural networks well-suited for image analysis tasks. For instance, Alirezazadeh et al. (2021) utilised CNNs to classify soil images into nine size classes but found that performance decreased with increasing image height. Azadnia et al. (2022) developed a CNN for classifying soil into 11 texture classes using smartphone pictures from varying heights, achieving the best performance with images taken from 20 cm. Additionally, Azizi et al. (2020) trained CNNs with stereo-pair images for classifying soil into six particle size classes, achieving a high classification accuracy of 98.72%. Swetha et al. (2020) created a smartphone application that uses random forest and CNNs to predict soil texture, achieving $R^2 \geqslant 96\%$ for clay and sand and $R^2 \geqslant 62\%$ for silt.

Despite these advancements, most CNN-based methods have focused on soil texture classification rather than predicting the entire PSD curve, which is essential for a comprehensive geotechnical analysis. Furthermore, existing approaches often fail to address the full complexity of soil types and conditions encountered in practical geotechnical engineering scenarios (Thien, 1979).

This study addresses these gaps by developing a novel CNN-based algorithm designed to predict the entire PSD curve from soil images, overcoming the limitations of traditional methods and existing image-based techniques. By employing MobileNet pre-trained on ImageNet with additional layers for transfer learning and capturing soil images under standardised lab conditions using a dark chamber with constant lighting, we aim to provide a robust, efficient and accurate method for PSD prediction. Our approach not only enhances the applicability of machine learning in geotechnical engineering but also offers a valuable tool for the characterization of a wide range of soil types in line with industry standards (ISO 14688–1, 2017; ISO 17892-4, 2016).

2. Materials and methods

In this study, soil samples primarily collected near Vienna (from the locations indicated by black dots in Fig. 1) are considered, for which PSDs were determined. Close-range photos of the soil samples and their PSDs are used as the inputs and outputs.

2.1. Data acquisition

2.1.1. Input data

The images are taken with smartphone cameras in a dark chamber to avoid light reflection (Fig. 2). The smartphone is placed on the top of the chamber and photos are taken from a 45 × 45 mm opening. The distance from the

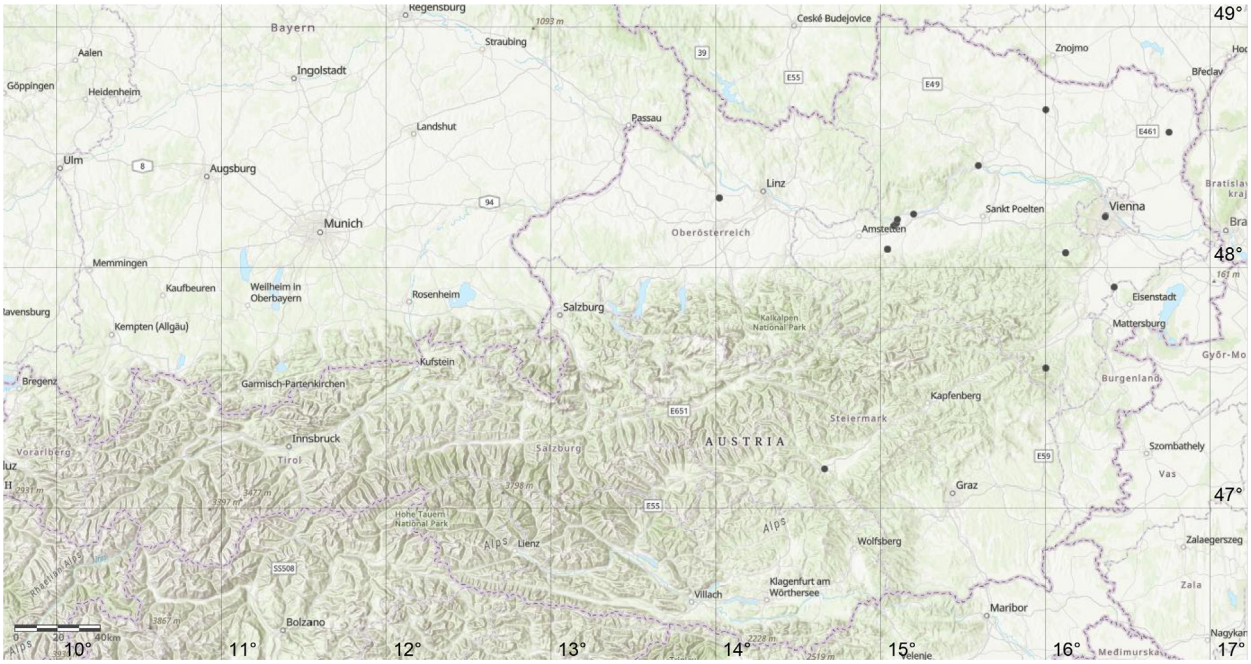


Fig. 1. GIS-visualisation (Redlands, 2024) of the locations of the soil samples retrieved (black dots). One additional sample was retrieved from Nepal.

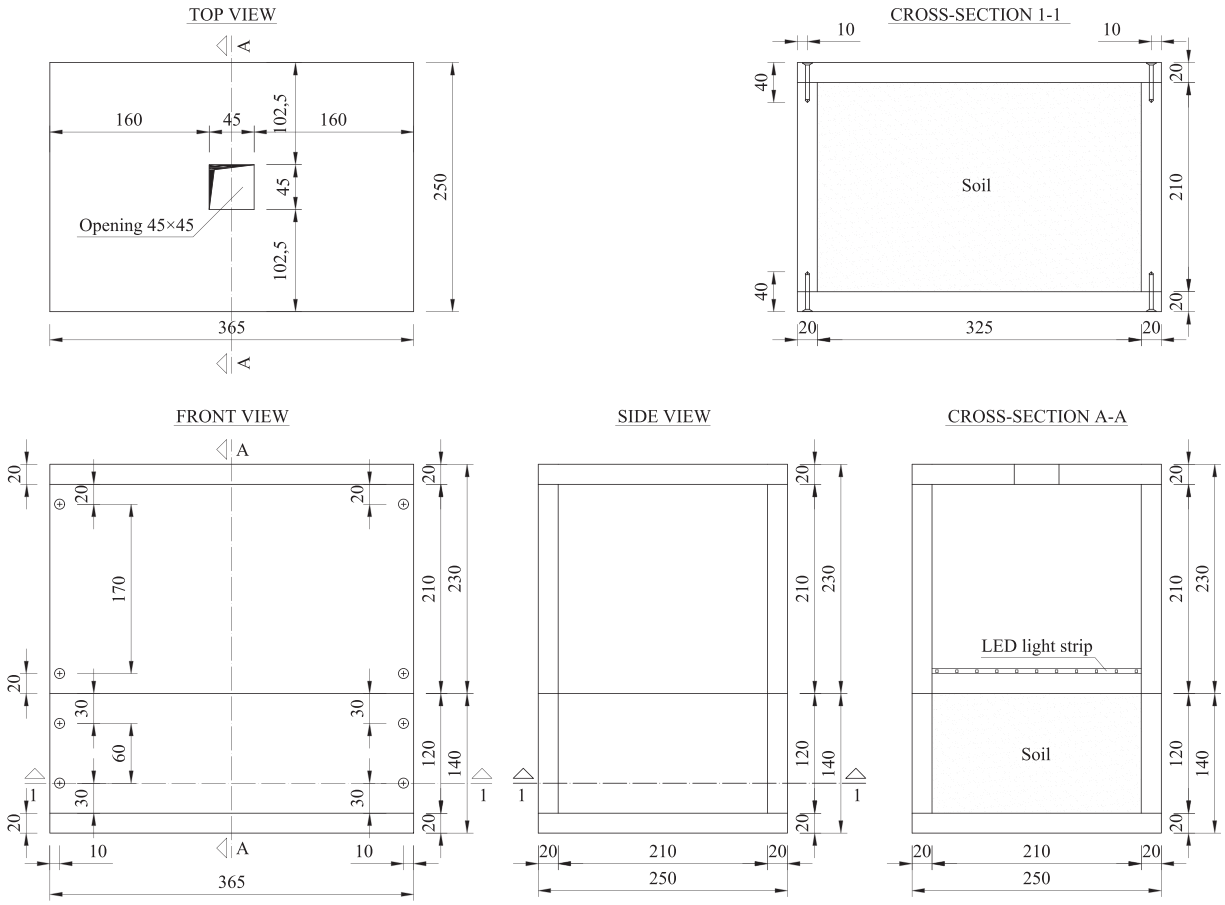


Fig. 2. Timber dark chamber for image acquisition.

soil surface to the lenses of the cameras is chosen as 210 mm based on a previous study (Azadnia et al., 2022). Since different smartphone cameras have different angles-of-view, different soil areas are photographed from the same distance. The inner ground dimensions are chosen under consideration of these areas i.e., trying to accommodate the area photographed by various smartphones from a 210 mm distance.

The soil samples are homogenised with the help of a cement mixer and oven-dried at 108°C. The soil is dry-pluviated (Raghunandan et al., 2012) from a height of 40 cm. Images are taken with the Motorola Edge and Samsung A52 smartphones with resolutions 4000×1800 and 6936×9248 pixels, respectively. One of the images taken for each soil is shown in Fig. 3. To avoid biases, more than one picture of the same soil is taken. Between the pictures, the soil surface is altered using various tools. In total, 136 pictures are taken. Pictures are imported as three-dimensional tensors, in which the first two dimensions correspond to the height and width and the third dimension to the colour channels.

2.1.2. Scaling

As each smartphone has its own resolution and physical dimensions of the photographed soil area, model generalisation must be ensured. Hence, all images are downsized to a common pixel density (pixel per mm = PPM), as shown in Table 1.

Moreover, since square tensors are generally preferred in CNNs due to the design simplicity of convolutional and pooling operations, as well as the prevalence of square images in many image datasets, the images are cropped into squares of dimensions ranging from 200×200 to 600×600 (Table 3).

2.1.3. Output data

The soil samples consist of silty, sandy and gravelly soils (Table 2). The PSD curves are obtained by sieving for the coarse and by the hydrometer sedimentation method (ISO 17892-4, 2016) for the fine fractions. The coefficients of uniformity and curvature of the curves vary between $C_U \in [1.9, 248.5]$ and $C_C \in [0.3, 49.4]$. It is therefore recommended to apply our method within these specified ranges.

In order to predict the entire PSD, the measurements are fitted with a function. In doing so, the prediction of the PSDs equals the prediction of their fitting parameters. Researchers attempted to fit the PSD with the log-normal distribution (Buchan et al., 1993), Gompertz and Fredlund curves (Fredlund et al., 2000; Hwang et al., 2002), fractal function (Bah et al., 2009), exponential and Weibull curves (Hwang, 2004). Although the Weibull distribution may not be well-suited for modelling multimodal distributions



Fig. 3. Smartphone photo image examples of all the soil samples.

Table 1
Image scaling.

Smartphone	Original images		PPM	Scaled images		PPM
	Width	Length		Width	Height	
Motorola Edge	1800	4000	11.5	720	1600	4.6
Samsung A52	6936	9248	26.1	1200	1600	4.6

(Zhang and Liu, 2006) or distributions with heavy tails (Mahbudi et al., 2020), it offers notable advantages in terms of high accuracy (as measured by R^2_{adj}, AIC and RMSE), flexibility (with applicability to various soil types) and simplicity (requiring only two to three parameters), as discussed by Esmaeelnejad et al. (2016). The general expression of the Weibull distribution is

$$F(d \leqslant d_i) = a - \exp\left[-\left(\frac{d_i}{b}\right)^c\right] \quad (1)$$

where d is the particle diameter finer than d_i; a, b and c are fitting parameters. For large particle diameters

$$\lim_{d\to\infty} F(d \leqslant d_i) = a \quad (2)$$

Moreover, since $F(d \leqslant d_i) = 100\%$ for large particle diameters, $a = 1$ and Eq. 1 becomes

$$F(d \leqslant d_i) = 1 - \exp\left[-\left(\frac{d_i}{b}\right)^c\right] \quad (3)$$

b shifts the fitting curve to the left (lower b) or right (higher b). The curves are steeper for higher c. The interpolated PSDs are shown in Fig. 4 and their parameters are listed in Table 2. The Weibull interpolation can introduce some level of error. However, even in cases where the interpolation visually appears less precise, it still yields a high coefficient of determination $R^2 \geqslant 0.983$. Therefore, the interpolation is a fairly accurate representation of the data.

Parameter b exhibits a wide range of values between 0.010 and 10.662. Hence, the natural logarithm of b is used as the output variable to stabilise the variance. Taking the natural logarithm compresses the range of values to $\ln(0.010) = -4.605$ and $\ln(10.662) = 2.367$.

Table 2
Soil sample identification number, provenance, Weibull fitting parameters, coefficient of determination of the fitting curves and soil fractions.

Id.	Provenance	Fitting parameters		R^2	Soil fractions				
		b (mm)	c (–)	(–)	Co (–)	Gr (–)	Sa (–)	Si (–)	Cl (–)
F827	Zistersdorf, AT	0.030	1.013	0.997	0.0%	0.0%	10.4%	80.1%	9.5%
G190	Aichdorf bei Judenburg	0.128	0.653	0.985	0.0%	9.0%	40.7%	45.1%	5.2%
H030	Wien, AT	0.165	0.737	0.996	0.0%	2.0%	60.7%	30.5%	6.8%
H031	St. Corona am Wechsel, AT	4.610	0.392	0.996	5.2%	44.3%	31.4%	17.2%	1.9%
H037	St. Corona am Wechsel, AT	1.935	0.392	0.991	0.0%	37.3%	34.9%	26.9%	0.9%
H038	St. Corona am Wechsel, AT	5.268	0.405	0.996	4.3%	48.4%	29.0%	16.5%	1.8%
H126	Wimpassing an der Leitha, AT	0.059	0.816	0.983	0.0%	5.2%	26.5%	62.0%	6.3%
H181	Ybbs, AT	4.856	0.829	0.999	0.0%	62.6%	33.1%	4.3%	0.0%
H183	Ybbs, AT	5.704	0.868	0.997	0.0%	67.7%	27.3%	4.9%	0.1%
H366	Steinakirchen, AT	7.549	0.610	0.990	0.0%	67.2%	23.3%	8.5%	1.0%
H367	Steinakirchen, AT	8.014	0.653	0.996	0.0%	69.6%	22.9%	6.7%	0.8%
H368	Steinakirchen, AT	9.242	0.817	0.995	0.0%	77.2%	17.8%	4.4%	0.6%
H371	Steinakirchen, AT	7.158	0.620	0.993	0.0%	67.0%	24.1%	7.8%	1.1%
H372	Steinakirchen, AT	7.253	0.663	0.990	0.0%	68.9%	22.4%	7.5%	1.2%
H374	Steinakirchen, AT	10.662	0.553	0.995	7.4%	63.2%	20.1%	8.2%	1.1%
H405	Hagsdorf, AT	0.041	0.599	0.996	0.0%	1.5%	22.8%	57.9%	17.8%
H493	Gajuri, NP	0.075	1.343	0.998	0.0%	0.8%	47.3%	48.7%	3.2%
H516	Omding, AT	2.472	1.107	0.998	0.0%	47.1%	52.1%	0.8%	0.0%
H549	Gottsdorf, AT	0.051	0.844	0.997	0.0%	0.2%	30.0%	64.9%	4.9%
H615	Pöchlarn, AT	7.949	0.885	0.993	0.0%	76.1%	19.0%	4.9%	0.0%
H616	Simbach, AT	0.245	3.284	0.998	0.0%	1.7%	95.5%	2.8%	0.0%
H617	Simbach, AT	0.152	0.986	0.996	0.0%	5.2%	60.1%	33.7%	1.0%
H637	Krems, AT	5.467	0.431	0.994	0.0%	53.1%	32.1%	12.8%	2.0%
H666	Heiligenkreuz, AT	0.093	1.493	0.998	0.0%	0.2%	53.0%	45.4%	1.4%
H668	Heiligenkreuz, AT	0.047	0.900	0.996	0.0%	3.0%	23.4%	67.0%	6.6%
H693	Hollabrun, AT	0.010	0.505	0.996	0.0%	0.0%	8.4%	56.4%	35.2%

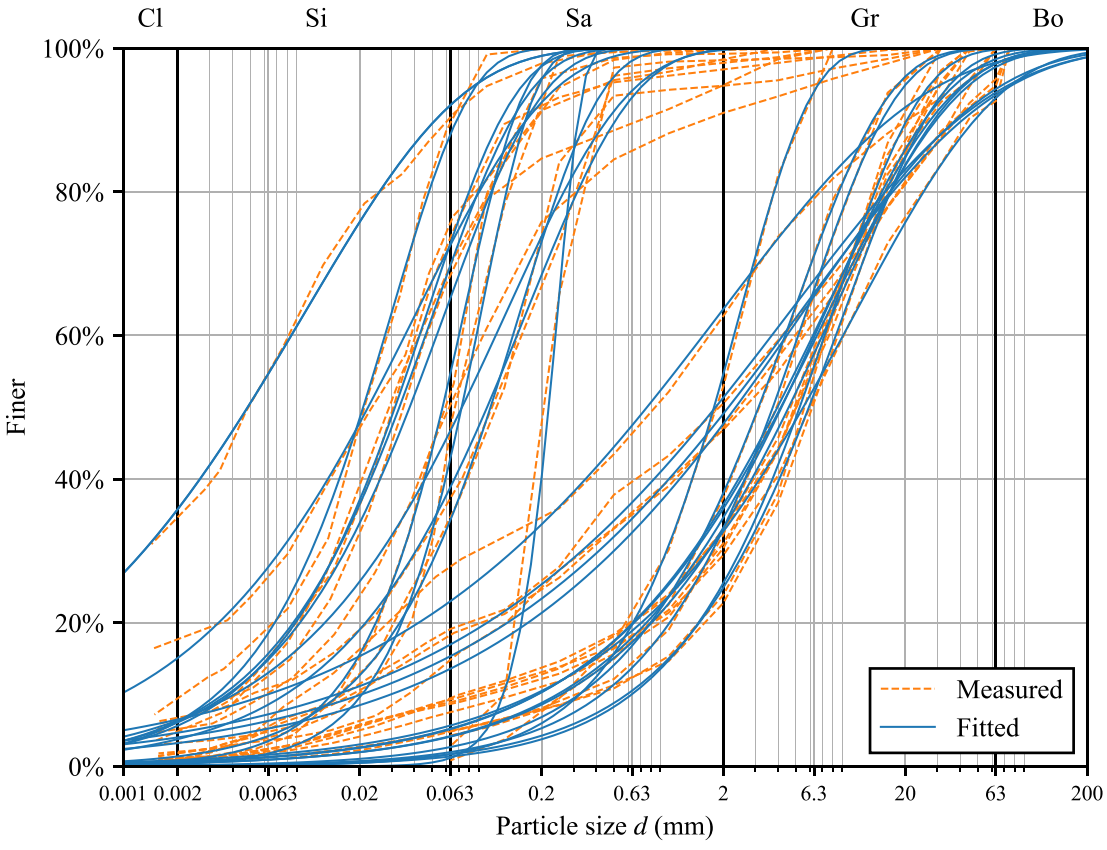


Fig. 4. PSDs of the training soils. Measured PSDs and Weibull interpolation.

The model's ability to predict the five particle size fractions outlined in Table 2, i.e. cobbles (Co), gravel (Gr), sand (Sa), silt (Si) and clay (Cl), has also been examined. In this context, the output vector encompasses the particle size fractions of each soil, collectively summing to one.

2.2. Convolutional neural network: mobilenet

CNNs (Fukushima, 1980) are widely used deep learning models with multiple layers of interconnected artificial neurons. Thy are designed to automatically learn and extract hierarchical representations from data to understand complex patterns and make accurate predictions. CNNs include convolutional layers that apply filters to input data, pooling layers that downsample the data and fully connected layers that perform regression or classification tasks. Convolutional layers can learn to detect various features, such as edges, textures or shapes, enabling the models to understand and classify images, videos and other structured data.

In traditional CNNs, each convolutional layer typically employs a set of kernels. Each kernel scans the entire image to extract features. For instance, in a 3×3 convolutional layer, a 3×3 kernel convolves across the entire image to detect various patterns. This process is performed for each feature map, resulting in a substantial computational load. In mathematical terms, convolution layers yield dot products B between the weights of the kernels K and subsets of the input tensors A according to Eq. 4 where n_K is the size of the kernels.

$$B_{ij} = \sum_{f=0}^{n_K-1} \sum_{h=0}^{n_K-1} A_{i+h,j+h} K_{i+f,j+h} \tag{4}$$

The kernels K slide one cell to the right until the end of the tensors A is reached, then down and right again. Hence, the dimensions of the tensors shrink by two at each convolution.

In this study, the MobileNet architecture (Howard et al., 2017) will be utilised, a lightweight and efficient CNN. MobileNet has been specifically designed for mobile deployment, offering a balance between model size and performance (Howard et al., 2017). This CNN is implemented by using the package `TensorFlow` (Abadi et al., 2015) in the Python 3.9 programming language (Van Rossum and Drake, 2009) and it is trained with GPU acceleration on a NVIDIA GeForce RTX 3060 graphic card through the `DirectML` package (White and Dykhuizen, 2022). In doing so, a 5× speed increase was achieved compared to benchmark CPU calculations. Although MobileNet is designed for efficient deployment on mobile devices, training the model on large datasets requires significant computational resources, which is why we utilised the RTX 3060 GPU for this process.

MobileNet's depth-wise separable convolution is a more efficient approach than traditional CNNs. Instead of using a single kernel to scan the entire input, the convolution process is divided into two steps: depth-wise and point-wise convolutions. In the first step, each input channel is convolved with its own separate kernel. Essentially, a set of 3×3 convolutions are performed on each input channel. Then, a 1×1 convolution is applied that combines the outputs from the previous step, resulting in a linear combination across all channels. This step captures global interactions between the local features.

Let us consider a simplified scenario with a RGB image as an example. In a conventional CNN layer, one might use 64 kernels, each with a size of $3 \times 3 \times 3$. This means $3 \times 3 \times 3$ convolutions are performed for each kernel across the entire image. In contrast, MobileNet begins with depth-wise convolution with 3 separate 3×3 kernels for each of the 3 input channels, resulting in 9 operations. Then, in the point-wise convolution, a single 1×1 kernel combines the outputs from these 9 operations. This final convolution is performed once across all channels. Each layer is followed by batch normalisation and ReLU non-linearity, except for the final fully connected layer.

The depth-wise convolution can be mathematically expressed as:

$$\widehat{G}_{k,l,m} = \sum_{i,j} \widehat{K}_{i,j,m} \cdot F_{k+i-1,l+j-1,m} \tag{5}$$

where $\widehat{K}$ is the depth-wise convolutional kernel of size $D_K \times D_K \times M$ and the m-th filter in $\widehat{K}$ is applied to the m-th channel in the feature map F to produce the m-th channel of the filtered output feature map $\widehat{G}$.

Collecting and annotating a large dataset of soil images with diverse characteristics is resource-intensive. Therefore, we start by loading the Keras implementation of the MobileNet CNN model (Fig. 5), whose weights are pre-trained on the ImageNet dataset (Deng et al., 2009). ImageNet is a large-scale dataset that consists of millions of labelled images commonly used for training computer vision models. Models trained on large and diverse datasets, learn to extract generic features from images that are often transferable and can be valuable for various image-related tasks. This technique, named "transfer learning" (Bozinovski, 2020), enables our model to leverage knowledge from one task to improve performance on a related task. The pre-trained layers in the base model are frozen to prevent their weights from being updated during training, while custom regression layers are added on top of the base model, thus allowing it to specialise. The output of the base model is passed through a global average pooling layer to aggregate spatial information. We adopt a single CNN with double regression outputs, represented by a `dense` layer with two nodes, to predict $\ln(b)$ and c. For the classification task, a `softmax` layer of size five is used to predict the particle size fractions (Fig. 6). Hence, two tasks, regression and classification, are employed in this study. The regression task focuses on predicting the parameters b and c of the Weibull distribution, which allows us to interpolate the entire PSD. The classification task is used to categorise

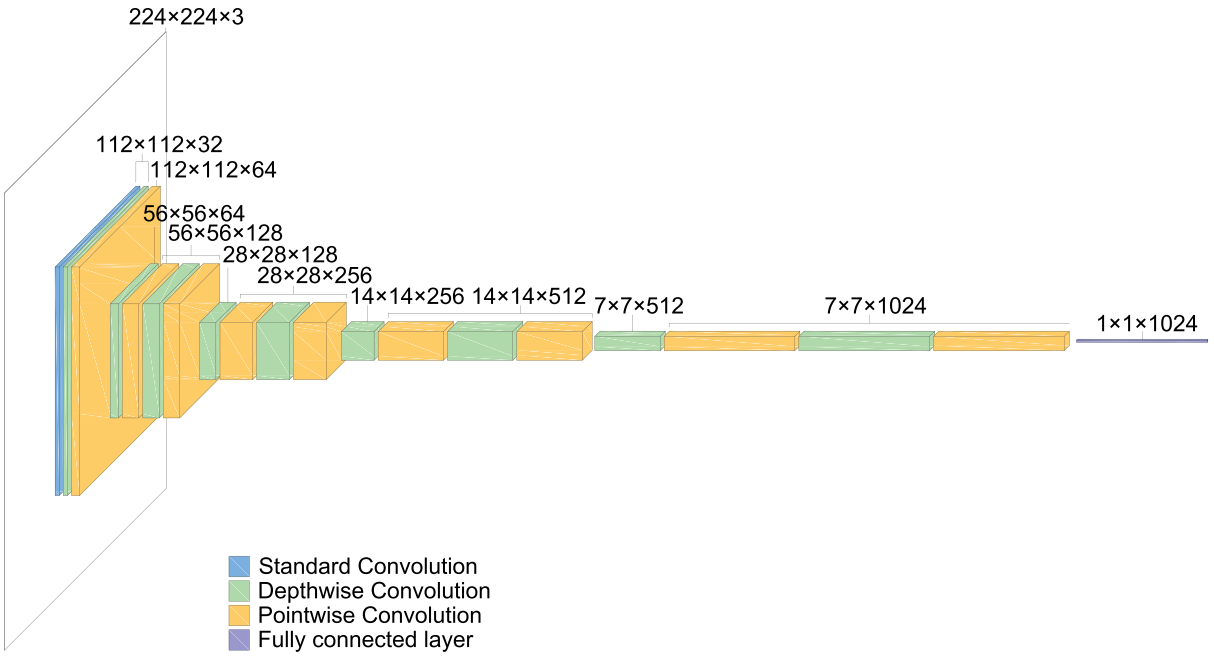


Fig. 5. Architecture of the MobileNet base model used in this study (based on the description of Howard, 2017).

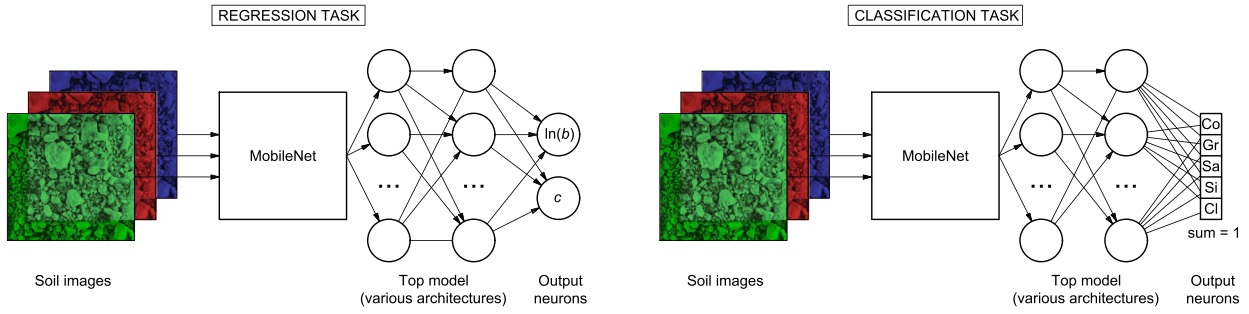


Fig. 6. Integration of the MobileNet base model with custom layers for both regression and classification tasks. The regression model predicts $\ln(b)$ and c, while the classification model categorises the soil into the five size fractions.

the soil based on its main particle fraction. This approach offers a quick and practical means of identifying the soil type, which can be useful for preliminary assessments.

2.2.1. Data splitting and augmentation

90% of the available images is used to train the CNN and 10% is assigned to testing. This split is performed by the function `train_test_split` of the `Scikit-learn` library (Pedregosa et al., 2011).

The training dataset size is increased through synthetic data augmentation, by applying the transformations and perturbations shown in Fig. 7.

2.2.2. CNN training

Depending on the number of additional layers and their neurons, the CNNs feature 4,280,514 to 17,031,490 parameters of which 1,051,650 to 13,802,626 are trainable. The non-trainable parameters are due to the pre-trained layers of MobilNet. The trainable parameters are optimised by minimising the mean squared error MSE of Eq. 6 ("back-propagation"), where y_i and $\hat{y}_i$ are the observed and predicted fitting parameters, respectively.

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{6}$$

MSE is minimised using the Adam method (Kingma and Ba, 2015), with an initial step size ("learning rate") of 0.001 and the parameters β_1 and β_2 controlling the decay rate of 0.9 and 0.999, respectively. Early stopping (Bengio, 2012) is implemented by monitoring the CNN performance to ensure that the MSE of the training set decreases of at least 0.001 after 50 steps ("patience"). This approach prevents overfitting, ensuring that the model generalises well to unseen data. Along with early stopping, we used model checkpointing to save the best-performing model based on test loss. This ensures that the final model used for evaluation is the one that demonstrated the best performance on the test set. The parameters of the CNN are updated after a mini-batch of data has passed through

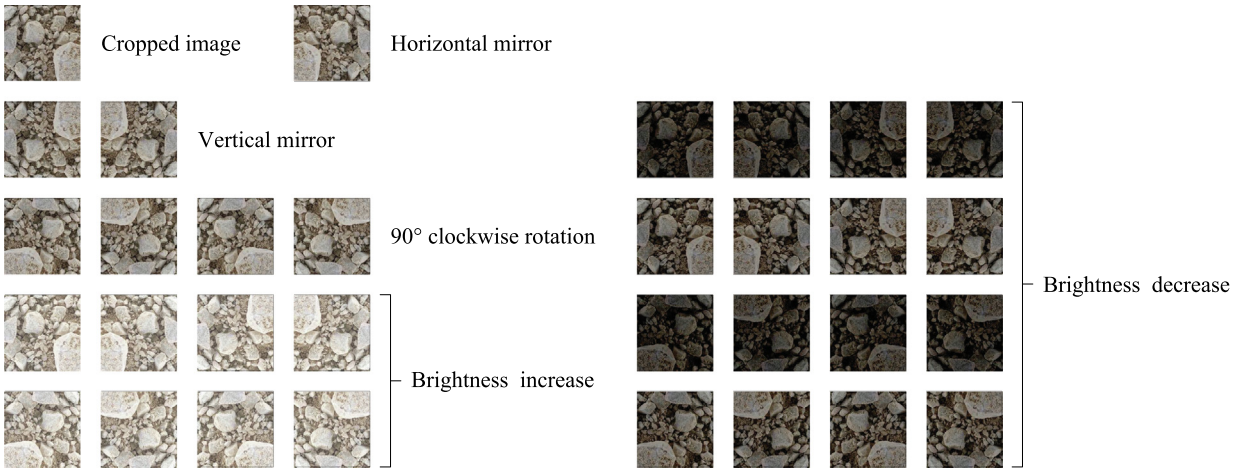


Fig. 7. Visualisation of the data augmentation techniques.

the network. The mini-batch size impacts the CNN training time (Bengio, 2012). Small mini-batches yield faster computations but require more iterations. There is no general agreement whether small (Hoffer et al., 2017) or large (Smith et al., 2018) mini-batches should be preferred. However, since larger batches determine faster iterations, the largest feasible mini-batch size will be preferred.

Once CNN training is completed, the coefficient of determination (R^2), root mean square error (RMSE), mean absolute error (MAE) and mean absolute percentage error (MAPE) are computed for the training and test sets according to the Eqs. (7)–(10) where $\bar{y}$ is the mean observed fitting parameter and n is the dataset size.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{7}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{8}$$

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{9}$$

$$\mathrm{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \tag{10}$$

The algorithm workflow is summarised in Fig. 8.

2.3. Test list

The tests in Table 3 have been conducted to optimise the hyperparameters. Firstly, different architectures are studied. Secondly, the optimal image size is determined in the range 200 × 200 to 600 × 600. This range is based on standard practices with CNNs. While MobileNet was designed for 224 × 224 images, the size can be adapted to specific

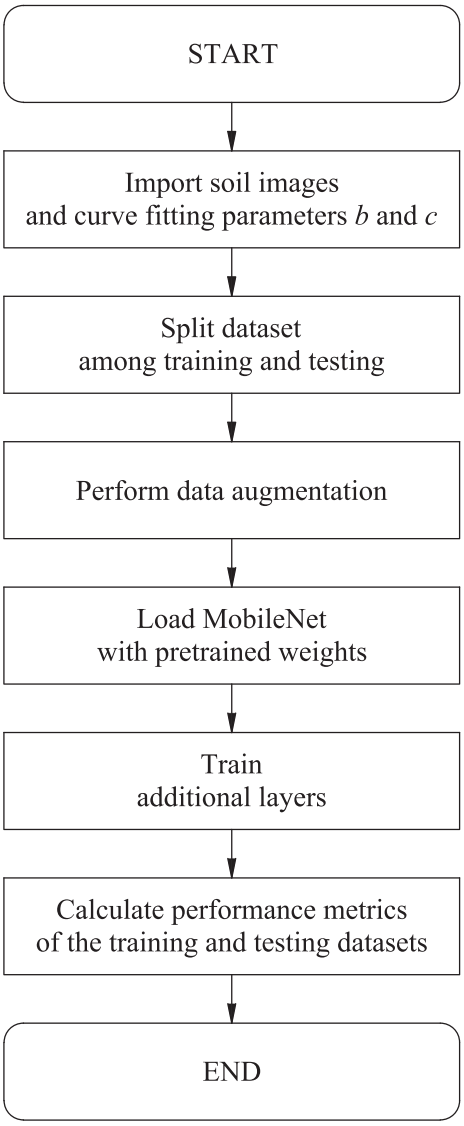


Fig. 8. Workflow of the algorithm.

Table 3
List of tests to optimise the hyperparameters.

No.	Neurons in trainable layers	Image size	Data augmentation*	Batch size	Task**
1	1024	200 × 200	HM, VM, R, BI, BD	1024	R
2.1	2048–512	200 × 200	HM, VM, R, BI, BD	1024	R
2.2	2048–512	300 × 300	HM, VM, R, BI, BD	512	R
2.3	2048–512	400 × 400	HM, VM, R, BI	512	R
3	2048–512–256	300 × 300	HM, VM, R, BI, BD	512	R
4.1	2048–512–256–128	300 × 300	HM, VM, R, BI, BD	512	R
4.2	2048–512–256–128	400 × 400	HM, VM, R, BI	512	R
4.3	2048–512–256–128	600 × 600	HM, VM, R	256	R
5.1	4096–2048–512–256–128	300 × 300	HM, VM, R, BI, BD	512	R
5.2	4096–2048–512–256–128	400 × 400	HM, VM, R, BI	512	R
5.3	4096–2048–512–256–128	600 × 600	HM, VM, R, BI	512	R
6	2048–512–256–128	400 × 400	HM, VM, R, BI	64	C

* HM: Horizontal mirror; VM: vertical mirror; R: 90° clockwise rotation; BI: Brightness increase; BD: Brightness decrease
** R: Regression; C: Classification

applications. This allowed us to strike a balance between efficiency and accuracy. Thirdly, since larger images require more memory, data augmentation is adapted accordingly, where all or just some of the image transformations are performed. The batch size is also impacted by memory requirements and thus adjusted.

Although the pseudorandom generator was set at a constant seed value, the model still exhibited some limited randomness, as a results of which, different CNN runs deliver slightly different results. This is due to GPU training and specifically to parallelism, memory management, concurrency and resource sharing. To mitigate this effect, the training and testing of the models are carried out five times and their results are averaged.

2.4. Explainable AI decision making

Explainable AI (XAI) aims to make AI models more interpretable by providing insights into how the models make predictions, allowing to understand and trust the reasoning behind the outputs.

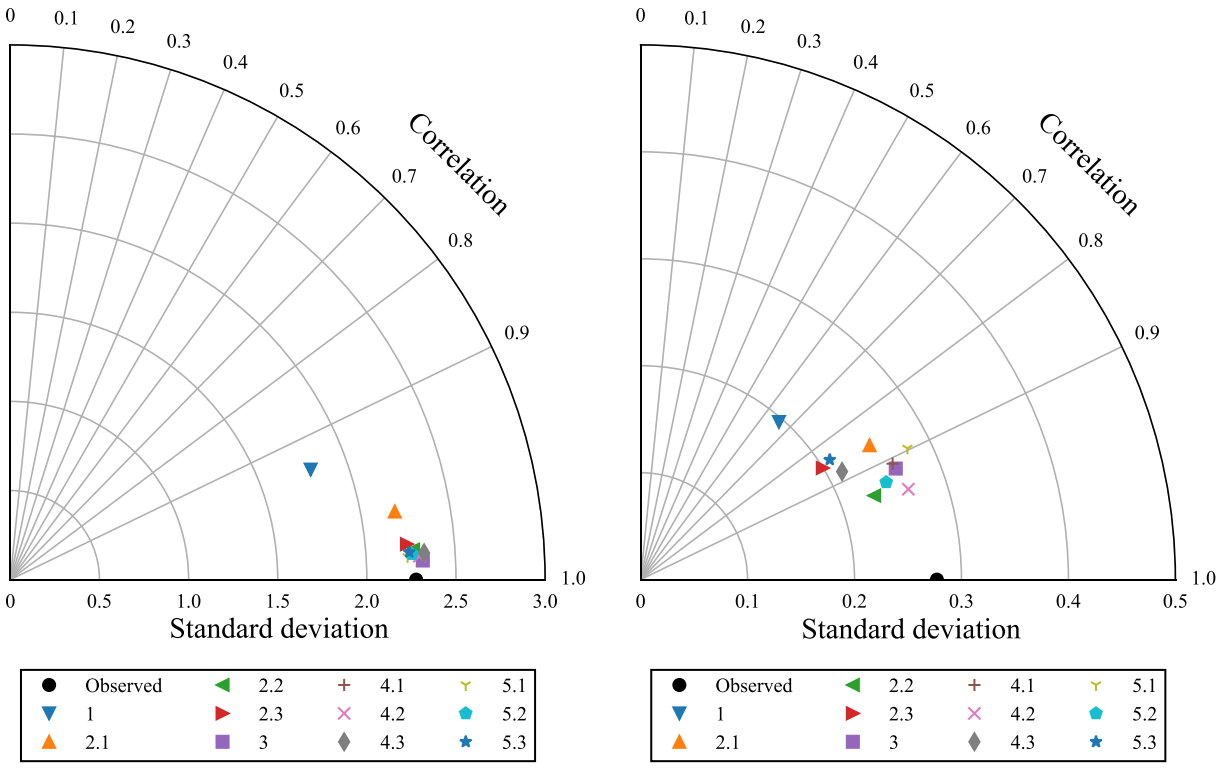
The importance of XAI arises from the adoption of AI in critical domains such as healthcare, finance and criminal justice. However, the importance of XAI stretches also to geotechnics, to build trust and confidence, making the technology more accountable and transparent.

There are several XAI methods for CNNs, such as the Gradient-weighted Class Activation Mapping (GradCAM) (Selvaraju et al., 2019) that highlights the important regions of an input image by leveraging the gradients of the target class with respect to the final convolutional layer of the model. GradCAM produces a heatmap that indicates the contribution of each pixel to the prediction. This approach has been applied to image classification or captioning, thus showing its potential for our application. In this study, the HiResCAM (Draelos and Carin, 2021) improved version of Grad-CAM is adopted. HiResCAM addresses a major limitation of Grad-CAM, which occasionally highlights incorrect locations due to gradient averaging. HiResCAM eliminates this step, ensuring that the relationship between the model's explanation and the prediction score is preserved.

HiResCAM typically operates on the final convolutional layer. To generate heatmaps, HiResCAM calculates gradients of the target class's score with respect to feature maps in this layer. The gradients are weighted by their global average pooling, producing importance scores for each feature map that represent the significance of each feature map on the prediction. HiResCAM combines the weighted

Table 4
Results of the regression tests 1–5. Mean values ± standard deviations

No.	$R^2_{test,b}$	$MAPE_{test,b}$	$R^2_{test,c}$	$MAPE_{test,c}$
1	0.833 ± 0.008	24.37% ± 1.56%	0.191 ± 0.237	23.99% ± 3.23%
2.1	0.948 ± 0.022	12.47% ± 3.36%	0.629 ± 0.099	16.05% ± 2.08%
2.2	0.988 ± 0.008	8.90% ± 2.55%	0.754 ± 0.112	12.07% ± 2.04%
2.3	0.932 ± 0.032	14.64% ± 3.99%	0.592 ± 0.061	14.06% ± 1.09%
3	0.997 ± 0.001	4.57% ± 0.48%	0.800 ± 0.054	11.08% ± 1.24%
4.1	0.996 ± 0.001	4.90% ± 0.34%	0.820 ± 0.067	10.64% ± 1.82%
4.2	0.995 ± 0.001	5.56% ± 0.93%	0.840 ± 0.052	10.75% ± 1.92%
4.3	0.991 ± 0.005	7.49% ± 1.37%	0.727 ± 0.086	15.00% ± 2.38%
5.1	0.996 ± 0.001	5.26% ± 0.50%	0.805 ± 0.021	11.53% ± 0.66%
5.2	0.995 ± 0.001	5.48% ± 0.72%	0.811 ± 0.042	11.80% ± 1.69%
5.3	0.992 ± 0.001	6.85% ± 0.75%	0.778 ± 0.096	12.13% ± 2.13%



(a) $\ln(b)$ (b) c

Fig. 9. Taylor diagram showing the performance of different models in predicting $\ln(b)$ and c. The diagram highlights the standard deviations and correlations of model predictions relative to observed values.

feature maps to create a single heatmap that visualises which parts of the input images are crucial for the prediction.

3. Results

The test results are shown in Table 4. They were conducted on a Windows machine and took approximately 20–80 min to complete. R^2 and MAPE are shown for $\ln(b)$ and c on the training and testing datasets. The model 4.2 achieves the highest R^2 score among the tested models, demonstrating the best performance for predicting soil parameters. This is confirmed by the Taylor diagram of Figure (where model 4.2. is the closest to the observations) 9 and by the Regression Error Characteristic (REC) curves of Fig. 10 (where model 4.2 shows the lowest Area Over the Curve or AOC, a measure of the expected error). Figs. 11 and 12 illustrate the model's predictions for $\ln(b)$ and c, respectively, compared to the interpolated values from Table 2. These figures show how well the model's predictions align with the expected values, with a shaded region representing a $\pm 30\%$ tolerance around the perfect prediction line. This visual comparison highlights the model's accuracy and reliability in predicting soil properties.

The image size 300×300 provided the best performance for the network of the test set 2. However, for the networks of the test sets 4 and 5, the best performance was obtained with the image size 400×400. According to the theory, more intricate architectures exhibit a greater appetite for data. If the right amount of data is provided, complex architectures deliver better performance, as confirmed by the performance increase between test sets 1, 2 and 3.

In the regression task where the observed values are non-zero, MAPE provides a meaningful percentage error measure. However, in the classification task, when dealing with a scenario where some classes or fractions might be zero (e.g., soil fractions with no gravel), MAPE becomes problematic because the denominator in the percentage error is zero, leading to errors. In contrast, MAE is a more appropriate metric for classification tasks as it measures the average absolute difference between predicted and observed values without dividing by the actual value, thus avoiding issues with zero denominators. Hence, the results of the classification task are shown in Table 5 with MAE as the chosen performance metric. As shown in Table 5, the model used in test 6 is very accurate in predicting all but the clayey soil fraction. In the latter case, R^2 is negative, implying that the performance is worse than a horizontal

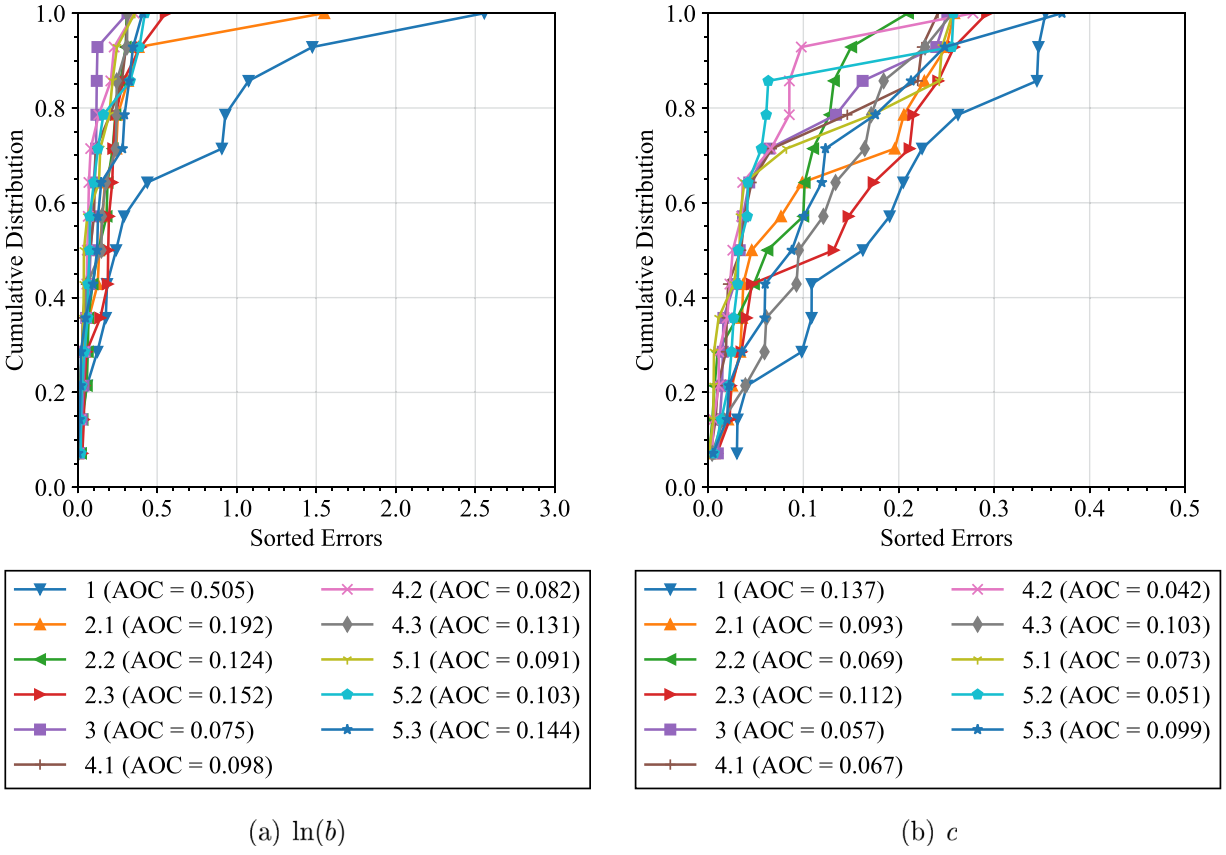


Fig. 10. REC curves showing the performance of different models in predicting ln(b) and c. Cumulative distribution of errors and AOC values indicated in the legend. Lower AOC values suggest better model performance.

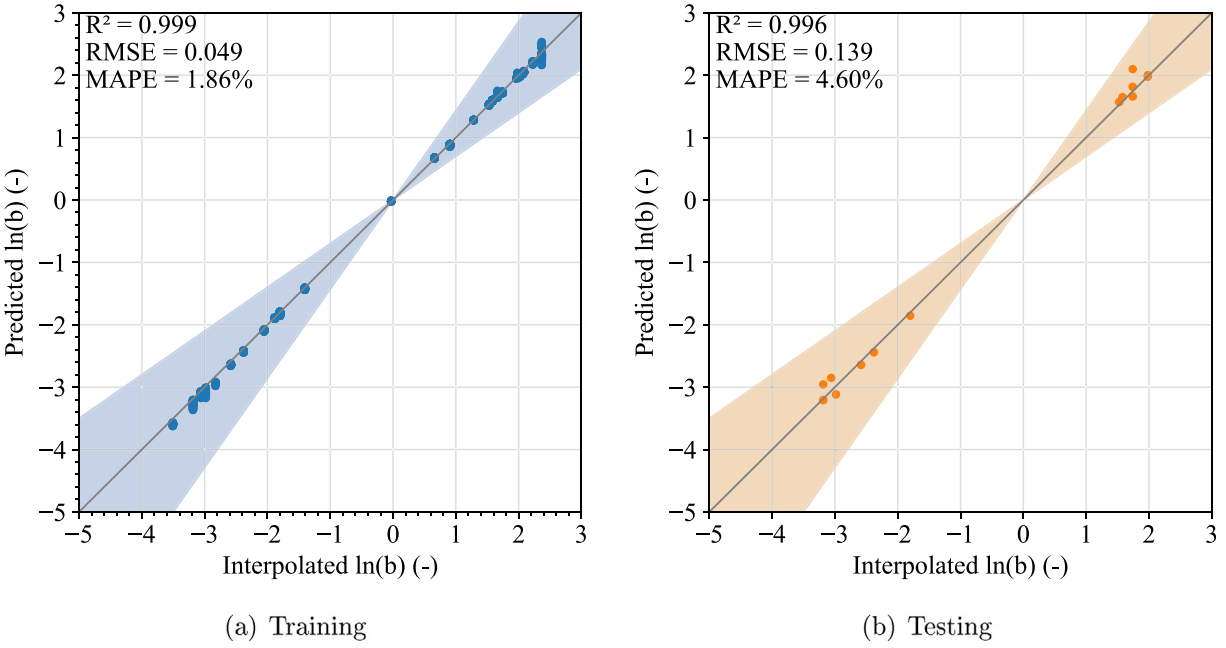


Fig. 11. Model 4.2 prediction of ln(b) against interpolated values from Table 2. Shaded region: ±30% distance from the perfect prediction.

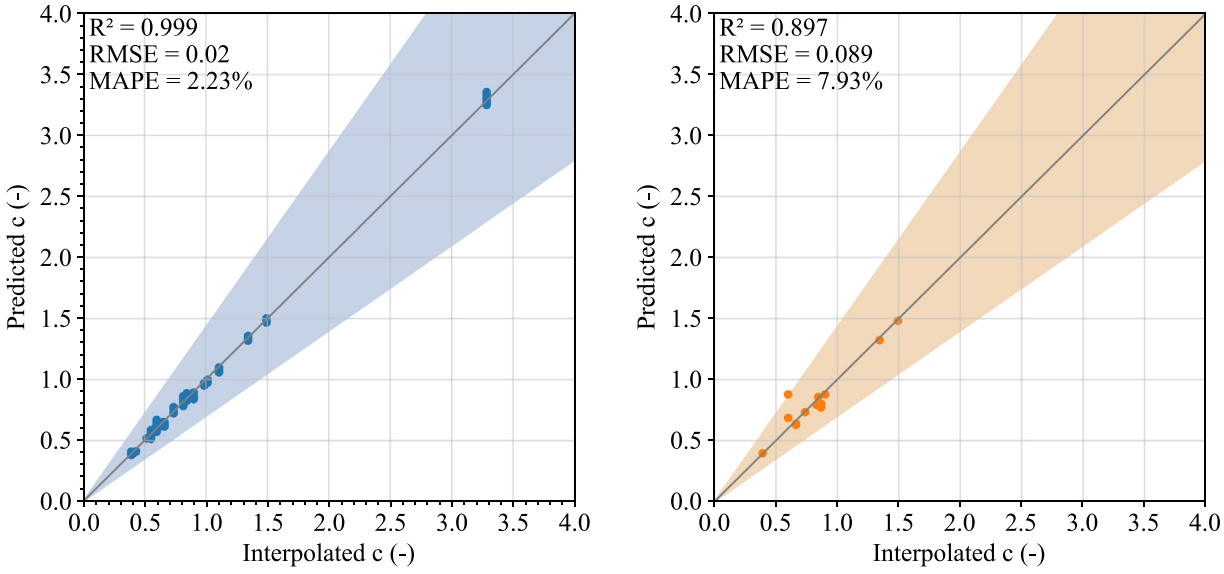


Fig. 12. Model 4.2 prediction of c against interpolated values from Table 2. Shaded region: ±30% distance from the perfect prediction.

Table 5
Results of the classification test 6. Mean values ± standard deviations.

Particle size fraction	R^2_{test}	MAE
Co	0.962 ± 0.037	0.17% ± 0.06%
Gr	0.992 ± 0.004	2.11% ± 0.77%
Sa	0.838 ± 0.030	3.95% ± 0.28%
Si	0.983 ± 0.011	2.22% ± 0.49%
Cl	0.682 ± 0.091	1.95% ± 0.34%

line, which would correspond to using the mean of the output as a predictor. Mathematically, this condition is obtained from Eq. 7 as $\sum_{i=1}^{n}(y_i-\hat{y}_i)^2 > \sum_{i=1}^{n}(y_i-\bar{y})^2$, suggesting that the model's errors are larger than the total variance of the output. (see Fig. 13).

The XAI heatmaps are superimposed on cropped images of the soil samples H616 and H637, as depicted in Fig. 14 and soil samples H031 and H374, as shown in Fig. 15. These heatmaps highlight in red the areas of the soil images that are most salient for the model's predictions of $\ln(b)l$ and c. For $\ln(b)$, the heatmaps often emphasise regions with larger particles, as this parameter is related to the mean particle size. In contrast, the heatmaps for c reveal a broader focus on diverse particle sizes and textures, reflecting its role in capturing the overall soil gradation. The differences observed between the heatmaps for $\ln(b)$ and c, particularly for H374, suggest that the model uses different image features for each parameter, with $\ln(b)$ being influenced by dominant grains and c by a more comprehensive analysis of soil texture and particle distribution.

4. Discussion

We have demonstrated that $\ln(b)$ and c can be fairly well predicted. However, we need to fully understand the implications of these predictions.

Firstly, we acknowledge the limitation imposed by the PPM value of 4.6 (Table 1), which restricts the ability to discern particle sizes smaller than $\frac{1}{4.6} = 0.22$ mm. However, our primary aim is not the direct identification of individual silt and clay particles; instead, it is the prediction of $\ln(b)$ and c to interpolate the entire PSD. The crux of our approach lies in the CNN's capability to deduce the overall shape of the curve, which is influenced by all particles within the soil sample. As highlighted in Table 4, our results are promising in predicting b and c. However, it is worth noting that our attempts to predict clay content alone (Table 5) did not yield satisfactory results, probably due to the fact that our dataset does not comprise purely clayey soils.

Secondly, what does the predictability of the fitting parameters mean? In essence, this implies our capability to generate values that closely aligns with the ones of the PSD interpolation. This gives rise to two questions: how closely does the resultant curve mirror the fitted curve and how accurately does it replicate the measured data? To answer these questions, we provide two illustrative examples (Fig. 16) based on the soil samples H183 and H493. Whereas the former shows relatively poor model performance (H183), the latter demonstrates excellent results (H493). When applied to H183, the model generates a curve that is slightly shifted to the right of the fitted

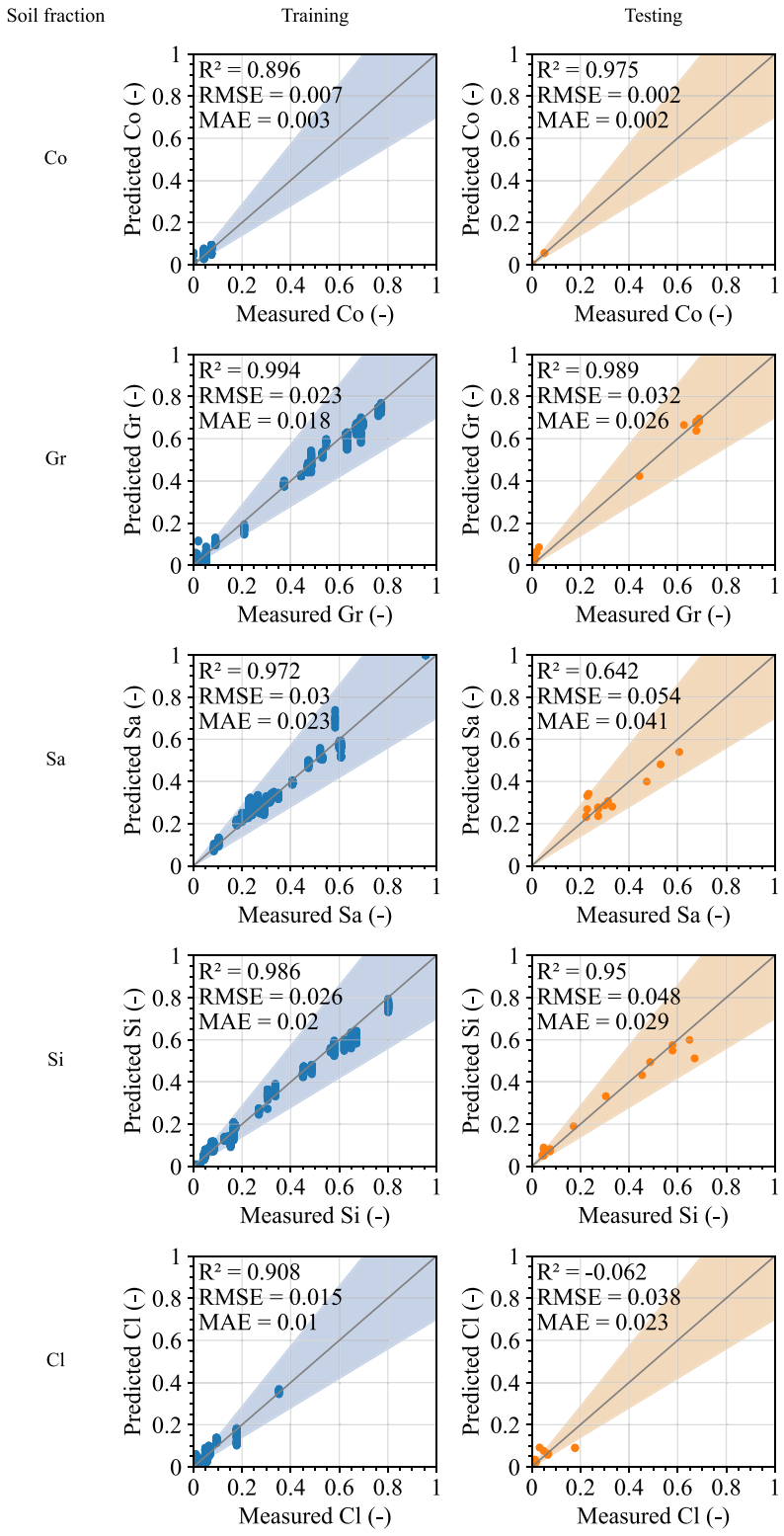
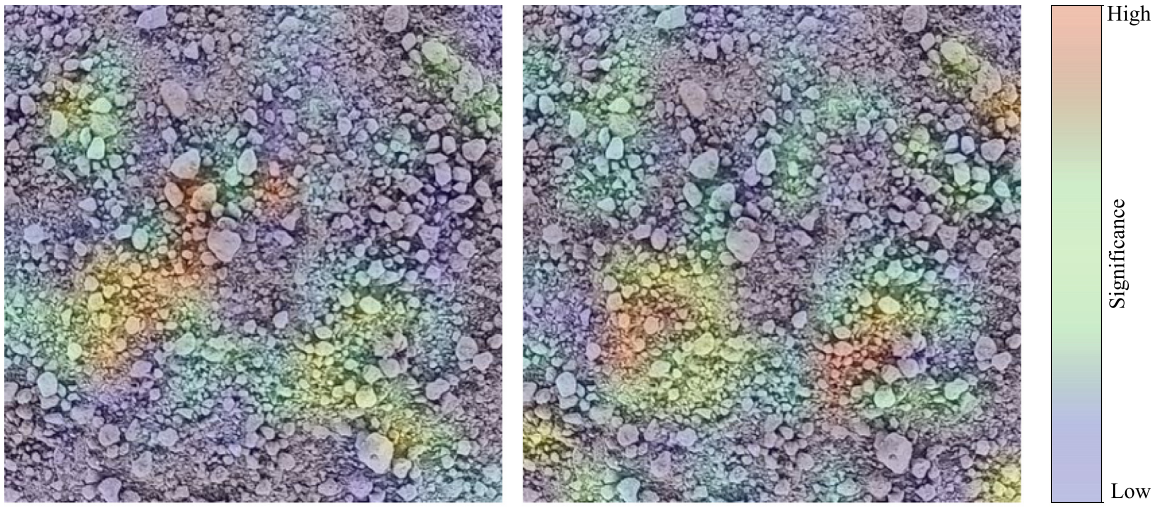


Fig. 13. Model 6 training and testing prediction of the soil fractions Co, Gr, Sa, Si and Cl. Shaded region: ±30% distance from the perfect prediction.

curve. However, the curve maintains the same overall shape and is not significantly distorted, the results being satisfactory, particularly for particle sizes in proximity of d_{10}. As for H493, the fitted curve exhibits remarkable agreement with the measurements, particularly for particle sizes larger than d_{10} and smaller than d_{90}. Nevertheless, it is worth noting that the observed deviations become noticeable primarily outside this range, reflecting the limitations of the curve fit.

Now we turn to the utilisation of XAI. By focusing on the parameter $\ln(b)$, which governs the horizontal position of the fitting curve, we can assume that the model examines



(a) Heatmap of $\ln(b)$ for soil H616 (b) Heatmap of c for soil H616



(d) Heatmap of $\ln(b)$ for soil H637 (e) Heatmap of c for soil H637

Fig. 14. HiResCAM heatmaps of the predictions of $\ln(b)$ and c for soils H616 and H637.

a region of the image that encompasses a characteristic particle size, such as d_{50}. On the other hand, when predicting parameter c, responsible for the steepness of the curve, the model should search for both large and small particle sizes. To substantiate this interpretation, we establish two mathematical correlations. Introducing d_{50} into Eq. 3, we obtain the relationship

$$0.50 = 1 - \exp\left[-\left(\frac{d_{50}}{b}\right)^{c}\right], \tag{11}$$

which can be rearranged to express d_{50} in terms of b:

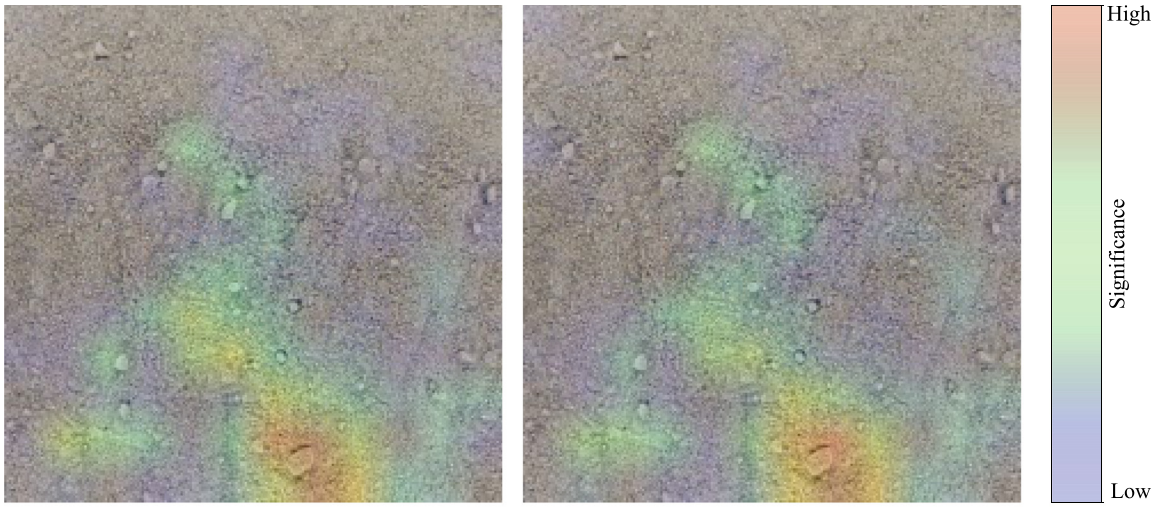
$$d_{50} = 0.69^{1/c} \cdot b \tag{12}$$

Since c is comprised in the interval $(0.392, 3.284)$, the term $0.69^{1/c}$ varies between 0.388 and 0.893. Hence, Eq. 12 is not very sensitive to c and confirms that, as d_{50} increases, so does b, providing evidence of a positive correlation. Similarly, by calculating the coefficient of uniformity, $C_U = \frac{d_{60}}{d_{10}}$, we find that

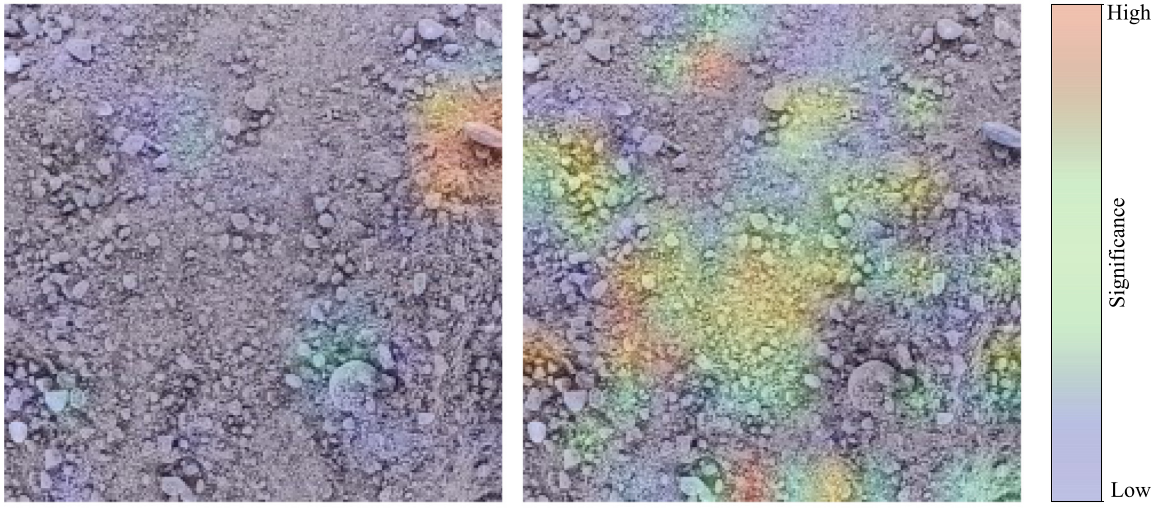
$$C_U = \frac{-b[\ln(0.40)]^{1/c}}{-b[\ln(0.90)]^{1/c}} \tag{13}$$

and thus $C_U = 8.70^{1/c}$. Hence, there is a negative correlation between C_U and c. These mathematical insights emphasise the interplay between the model's predictions and the characteristics of the soil particles.

According to the standards (ISO 14688-2, 2017), one of the soils in Fig. 14 is "uniformly graded" (H616) with $C_U = 1.9$, while the other is "well graded" (H637) with $C_U = 151.3$. The heatmaps of the two soils exhibit slightly distinct behaviours. Specifically, for both soils, the model estimates both output parameters from a similar image region. However, this effect is more pronounced when the soil is well graded (H637). One possible interpretation is



(a) Heatmap of $\ln(b)$ for soil H031 (b) Heatmap of c for soil H031



(d) Heatmap of $\ln(b)$ for soil H374 (e) Heatmap of c for soil H374

Fig. 15. HiResCAM heatmaps of the predictions of $\ln(b)$ and c for soils H031 and H374.

that the model estimates b from a region that represents the mean particle size, whereas for c, both small and large particle sizes are required to determine whether the soil is uniformly graded. Consequently, in well-graded soils like H637, where various particle sizes are found around the mean diameter, the model tends to focus on the same image area for both parameters.

It is also worth noting in Fig. 15 that the CNN tends to focus on individual grains that are significantly larger than the rest. This observation suggests that these outliers play a crucial role in the prediction of $\ln(b)$ and c. Comparing Figs. 15, c–d, it becomes apparent that these larger grains consistently hold significance for $\ln(b)$. However, the same level of significance for c is not observed across all soils, as it appears to be only relevant for the soils with a better-graded distribution (H031). This underscores the CNN's ability to discern the importance of individual grains within the context of different soil types and gradings.

5. Conclusions

This study presents a CNN for predicting the PSD based on smartphone images taken from a dark chamber. The main advantage of our model is the ability to provide rapid and efficient predictions. This real-time capability is important in geotechnical engineering, where quick and accurate assessment of soil properties is essential. To support the claim of real-time prediction capabilities, we have deployed the proposed model online via the Hugging Face platform, accessible at https://grai.boku.ac.at. This platform allows users to upload soil images and receive PSD predictions in real-time. The use of a mobile phone for photo capture underscores the practical and accessible nature of the system, enabling field use where quick and accurate soil assessments are needed. This real-time capability provides significant advantages in geotechnical engineering for immediate decision-making and field evaluations.

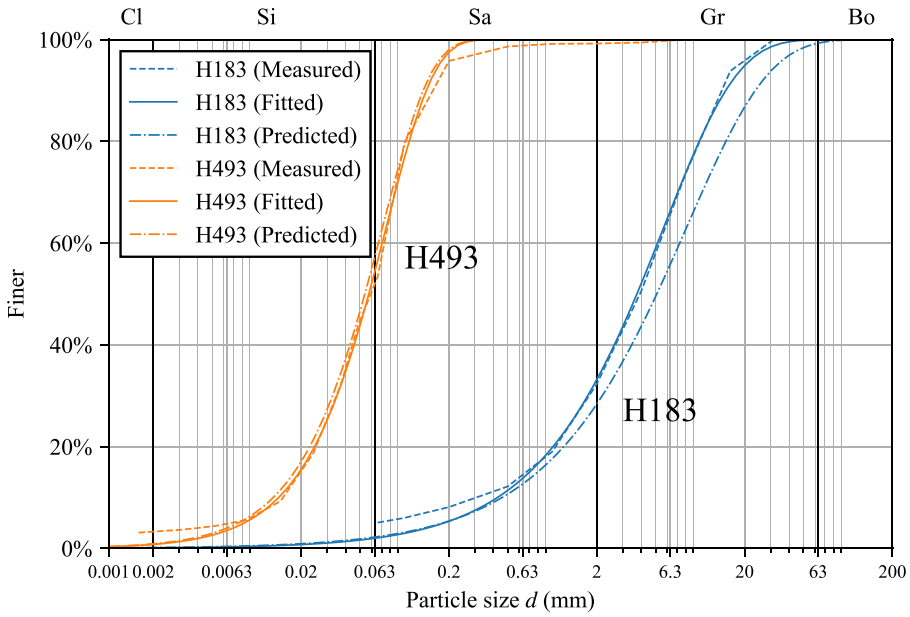


Fig. 16. Visual appraisal of the model prediction of the PSD curve for H183 and H493.

To enhance model's robustness, we tested a wide spectrum of soils, from clayey silt to gravel. The diverse soils show the reliability and relevance of our predictions. However, since purely clayey soils are absent in our analysis, future research should prioritise their inclusion. Moreover, the dataset originates from a specific geographic region. Further studies should consider samples from other locations. It is worth noting that, despite employing data augmentation techniques for input features, the output labels remain fairly consistent. Additionally, in our study, we employed two smartphones for capturing images. To improve generalisability, it is essential to integrate a more extensive range of smartphones with different camera specifications.

The huge data volume also poses some challenges. To overcome this, several strategies can be explored, such as using cloud services, supercomputers and batch processing. Batch processing involves breaking down large datasets into smaller, manageable segments, enabling to overcome the resources' constraints.

In addition to these limitations, the Weibull interpolation exhibits diminished efficacy at the tails below d_{10} and above d_{90}. Hence, subtle challenges may arise in these regions.

The method in this study could have limitations when applied to soils with different properties. Fine, organic-rich, aggregated and wet soils can all impact the method's accuracy. We suggest calibrating the model for different soil conditions and incorporating additional soil property analyses to improve accuracy.

Despite these limitations, a significant advancement of our study lies in the adoption of the geotechnical classification approach. Instead of predicting soil texture or discrete particle size fractions, we succeeded in predicting the entire PSD curve, offering a comprehensive and continuous characterization of soil gradation. This novel method surpasses traditional discrete classification by providing a full PSD curve, which is crucial for accurate geotechnical analysis.

Moreover, we employed XAI techniques, specifically the HiResCAM method, to gain insights into the decision-making process of our model. This approach not only reveals which regions of the soil images contribute most to the predictions of $\ln(b)$ and c, but also enhances transparency and trust in the model's outputs.

Our research also explores the use of smartphone technology for field applications, demonstrating that high-quality soil analysis can be achieved with accessible and cost-effective tools. This aspect of the study highlights the practical implications of our approach for real-world geotechnical assessments.

In summary, our study presents a novel CNN model for real-time PSD prediction and advances the field of soil characterization through a combination of innovative methodologies and practical applications.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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